

Class #7 - Applications to Differential Equations

1. Consider a _____linear system of DEs:

$$\frac{dx}{dt} = Ax, \quad x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \quad (*)$$

Goal: to find _____linearly _____solutions $x^1(t), \dots, x^n(t)$.

2. For each eigenvector v^j of A with eigenvalue λ_j , we have a solution $x^j(t) = \underline{\hspace{2cm}}$ of (*). If A has n linearly independent eigenvectors $v^1, \dots, v^n$ with eigenvalues $\lambda_1, \dots, \lambda_n$ respectively ($\lambda_1, \dots, \lambda_n$ need _____be distinct), then $x^j(t) = \underline{\hspace{2cm}}$, $j = 1, \dots, n$ are _____linearly independent solutions of (*).

3. Example: Let

$$A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}.$$

The characteristic polynomial $p(\lambda) = \underline{\hspace{2cm}} = (1 - \lambda)(\lambda - 3)(\lambda + 2)$.

Eigenvalues of A : _____

corresponding eigenvectors:

$$v^1 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}, \quad v^2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad v^3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

General solution is